

A Confidence-Guided Framework for Culturally Aware Image Captioning Using Lightweight Recurrent Models

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ABSTRACT Image captioning systems based on vision–language models often generate culturally inappropriate or hallucinated descriptions when applied across diverse social and regional contexts. To address this limitation, this paper proposes a confidence-guided culturally aware image captioning framework that explicitly regulates cultural enrichment rather than embedding implicit cultural assumptions. The system first produces a visually grounded base caption using a pretrained vision–language model and then evaluates cultural relevance through two lightweight classifiers: an Echo State Network and a Gated Recurrent Unit network. Cultural context is injected into the caption only when the predicted confidence exceeds a predefined threshold, thereby suppressing hallucinations under uncertainty. The framework further integrates multilingual refinement, cultural entity extraction, and controlled stylistic adaptation within a unified pipeline. Experimental results show that the Gated Recurrent Unit achieves higher cultural classification accuracy, while the Echo State Network provides faster and more interpretable inference with lower computational cost. Confidence-based decision fusion reduces cultural hallucination and improves suppression accuracy without degrading standard caption quality metrics, demonstrating reliable and deployable culturally aware image captioning.

INDEX TERMS Culturally aware captioning, confidence-guided learning, echo state networks, image captioning, multilingual vision–language systems

I. INTRODUCTION

RECENT advances in computer vision and multimodal learning have enabled image captioning systems to generate fluent natural language descriptions from visual content. Such systems play a critical role in applications including accessibility for visually impaired users, digital content management, social media automation, cultural heritage preservation, and human–computer interaction. With the widespread deployment of vision–language models across global and culturally diverse user bases, the need for captions that are not only visually accurate but also culturally sensitive and context-aware has become increasingly important. Conventional image captioning approaches, however, often prioritize syntactic correctness and object recognition while overlooking deeper cultural interpretation.

State-of-the-art image captioning systems are predominantly based on large transformer architectures and extensive multimodal pretraining. While these models achieve high performance on standard linguistic metrics, they frequently

suffer from cultural hallucinations, overgeneralization, and a lack of interpretability, particularly when describing culturally ambiguous scenes such as traditional food, festivals, rituals, or everyday practices. Moreover, their large computational footprint and opaque decision-making processes limit their deployment in resource-constrained environments and safety-critical applications. These challenges highlight the need for lightweight, explainable mechanisms that can guide cultural enrichment without introducing spurious or insensitive content.

In response to these limitations, this paper proposes a culturally aware and confidence-guided image captioning framework that augments base visual descriptions with controlled cultural reasoning. The system combines a pretrained vision–language captioning model with two complementary cultural classifiers: an Echo State Network (ESN) for fast, interpretable inference and a Gated Recurrent Unit (GRU) for higher classification accuracy. Cultural enrichment is injected only when model confidence exceeds a predefined

threshold, thereby suppressing hallucinations in uncertain cases. The framework further integrates multilingual translation, hashtag generation, cultural entity extraction, interactive question–answering, and cultural memory retrieval, forming a unified and deployable system.

The remainder of this paper is organized as follows. Section II reviews related work in image captioning, cultural AI, and hallucination mitigation. Section III describes the proposed system architecture and methodology in detail. Section IV discusses the results and provides a comparative analysis of the ESN and GRU models. Finally, Section V concludes the paper and outlines directions for future research.

II. LITERATURE REVIEW

Image captioning is a fundamental problem in multimodal artificial intelligence that aims to generate coherent natural language descriptions for visual content. Early deep learning approaches primarily adopted encoder–decoder frameworks, where convolutional neural networks (CNNs) extracted visual features and recurrent neural networks (RNNs), such as LSTMs or GRUs, generated textual sequences. Surveys by Hrga and Ivacic-Kos [2] and Stefanini et al. [7] provide comprehensive overviews of these architectures, highlighting the role of attention mechanisms in aligning image regions with generated words. Although these methods achieved strong performance on benchmark datasets, their optimization objectives largely focused on syntactic fluency and object-level accuracy, offering limited sensitivity to cultural or contextual nuances.

With the emergence of transformer architectures, image captioning systems experienced a paradigm shift toward joint vision–language modeling. Multimodal transformers and cross-attention mechanisms enabled more expressive alignment between visual representations and textual tokens, as demonstrated in works such as [14] and [21]. Large-scale pretraining on paired image–text corpora further improved caption quality and generalization. However, these gains came at the cost of significantly increased computational complexity, reduced interpretability, and a tendency to hallucinate semantically plausible but visually unsupported details, particularly when cultural cues are weak or ambiguous.

Multilingual and cross-lingual image captioning has received growing attention as systems are deployed globally. Datasets such as AraTraditions10k [1] and ArtELingo-28 [15] explicitly address linguistic and cultural diversity by providing captions across multiple languages and cultural domains. Retrieval-augmented and prompt-based approaches, including LMCap [12], leverage large language models to enable few-shot multilingual captioning. Regional systems targeting Indian languages [5], [19] and Arabic captioning frameworks [4], [11] further extend accessibility through multilingual and voice-based interfaces. Despite these advancements, most multilingual approaches implicitly assume that accurate translation ensures cultural appropriateness, an assumption that often fails when cultural practices, rituals, or food traditions require localized interpretation beyond literal

semantics.

Several recent studies have attempted to incorporate cultural and heritage awareness directly into captioning pipelines. Cultural heritage–oriented approaches employ curated datasets or augmentation strategies to improve representation of traditional artifacts, festivals, and rituals [3]. Region-specific captioning systems, such as Thai cultural image captioning [8], demonstrate that incorporating localized visual and linguistic priors improves descriptive authenticity. However, these models are typically domain-specific and dataset-bound, limiting their generalization to unseen cultural contexts or mixed scenes. Moreover, cultural enrichment is often applied uniformly, without considering uncertainty in visual evidence.

Context-aware attention mechanisms and knowledge-augmented captioning models have also been explored to enhance semantic richness. Approaches integrating external commonsense or cultural knowledge [10] and multi-agent multimodal reasoning frameworks [20] enable more expressive and informative captions. While effective, these systems introduce additional sources of hallucination when external knowledge is weakly grounded in visual input. Their complexity also increases inference cost and reduces transparency, making them difficult to deploy in real-time or resource-constrained environments.

Bias, hallucination, and expressive variability have emerged as critical challenges in vision–language models. Studies such as [13] analyze multimodal bias and emotional context leakage, while [18] highlights significant variation in semantic framing across languages. These works underscore the risk of culturally inappropriate or overconfident captions, particularly when models extrapolate beyond visual evidence. However, most mitigation strategies focus on post-hoc analysis or language-level correction rather than architectural mechanisms that explicitly regulate cultural inference.

Overall, existing image captioning research demonstrates strong progress in linguistic fluency, multilinguality, and semantic alignment, yet lacks explicit mechanisms for confidence-aware cultural reasoning. Cultural enrichment is often implicit, opaque, and prone to hallucination, while lightweight and interpretable alternatives remain underexplored. These limitations motivate the need for captioning systems that can balance descriptive richness with cultural appropriateness, selectively inject cultural context based on confidence, and operate efficiently in real-world deployment scenarios. Table 1 summarizes the key limitations of existing image captioning approaches, highlighting gaps in cultural awareness, confidence-driven hallucination control, interpretability, and lightweight deployable modeling that motivate the proposed system.

III. METHODOLOGY

This section presents the proposed confidence-guided culturally aware image captioning methodology. The framework integrates pretrained vision–language models with lightweight recurrent cultural classifiers and explicitly reg-

TABLE 1: Research Gaps in Existing Image Captioning Literature

Aspect	Identified Research Gap
Cultural Awareness	Most captioning models emphasize object recognition and fluency, with limited cultural grounding [2], [7].
Multilingual Captioning	Language translation is prioritized over cultural appropriateness [1], [12].
Hallucination Control	Cultural hallucinations are not explicitly suppressed under visual ambiguity [13].
Interpretability	Transformer and multi-agent models lack transparent cultural reasoning [20].
Computational Cost	Heavy reliance on large pretrained models limits real-time deployment [21].
Generalization	Cultural models are often dataset- or region-specific [8].
Confidence Awareness	Prior work does not modulate cultural enrichment based on model confidence.
Lightweight Modeling	Limited exploration of efficient recurrent or reservoir-based cultural reasoning models.

ulates cultural enrichment using probabilistic confidence thresholds. The overall objective is to generate visually grounded, culturally appropriate, and hallucination-controlled captions while maintaining computational efficiency and interpretability.

A. PROBLEM FORMULATION

Let an input image be denoted as

$$I \in \mathbb{R}^{H \times W \times 3}. \quad (1)$$

The goal of image captioning is to generate a natural language sequence

$$C = \{w_1, w_2, \dots, w_T\}, \quad (2)$$

where each token w_t belongs to a vocabulary $\mathcal{V}$, such that C accurately describes the visual content of I .

In addition to visual correctness, the proposed system determines whether cultural context should be injected. Let the cultural mode variable be defined as

$$y \in \mathcal{Y} = \{\text{food_traditional, festival_context, daily_life, generic}\}. \quad (3)$$

The task is thus extended to jointly estimate

$$P(C | I), \quad (4)$$

and

$$P(y | C), \quad (5)$$

while injecting cultural information only when the confidence of $P(y | C)$ exceeds a predefined threshold.

B. BASE CAPTION GENERATION

A pretrained vision–language model generates the initial caption. Given image I , a visual encoder extracts dense visual features

$$\mathbf{v} = f_{\text{vis}}(I), \quad (6)$$

where $f_{\text{vis}}(\cdot)$ denotes the vision backbone.

A causal language decoder then autoregressively generates the base caption

$$P(C | I) = \prod_{t=1}^T P(w_t | w_{<t}, \mathbf{v}). \quad (7)$$

This stage prioritizes visual fidelity and linguistic fluency, producing a visually grounded but culturally neutral caption C_{base} .

C. CULTURAL FEATURE REPRESENTATION

The generated caption C_{base} is converted into a numerical feature vector

$$\mathbf{x} = \phi(C_{\text{base}}) \in \mathbb{R}^d, \quad (8)$$

where $\phi(\cdot)$ encodes lexical, structural, and semantic indicators such as caption length, adjective density, ritual keywords, utensil presence, sentiment polarity, and token diversity.

These features serve as inputs to two parallel cultural classifiers: an Echo State Network (ESN) and a Gated Recurrent Unit (GRU).

D. ECHO STATE NETWORK FOR CULTURAL CLASSIFICATION

The ESN consists of a fixed recurrent reservoir with randomly initialized weights. The reservoir state update is given by

$$\mathbf{h}_t = (1 - \alpha)\mathbf{h}_{t-1} + \alpha \tanh(\mathbf{W}_{\text{in}}\mathbf{x}_t + \mathbf{W}\mathbf{h}_{t-1}), \quad (9)$$

where α is the leak rate and $\mathbf{W}$ is constrained by spectral radius $\rho < 1$.

Only the output weights are trained using ridge regression

$$\mathbf{W}_{\text{out}} = \arg \min_{\mathbf{W}} \|\mathbf{H}\mathbf{W} - \mathbf{Y}\|^2 + \lambda \|\mathbf{W}\|^2, \quad (10)$$

where $\mathbf{H}$ is the reservoir state matrix and $\mathbf{Y}$ is the one-hot encoded label matrix.

The ESN confidence is computed as

$$p_{\text{ESN}} = \max(\text{softmax}(\mathbf{W}_{\text{out}}^\top \mathbf{h})). \quad (11)$$

E. GRU-BASED CULTURAL CLASSIFIER

The GRU update equations are

$$\mathbf{z}_t = \sigma(\mathbf{W}_z\mathbf{x}_t + \mathbf{U}_z\mathbf{h}_{t-1}), \quad (12)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r\mathbf{x}_t + \mathbf{U}_r\mathbf{h}_{t-1}), \quad (13)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h\mathbf{x}_t + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{h}_{t-1})), \quad (14)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \quad (15)$$

Algorithm 1 Echo State Network for Cultural Classification**Require:** Feature vector $\mathbf{x}$ **Ensure:** Predicted class y , confidence p

- 1: Initialize reservoir state $\mathbf{h}_0 = \mathbf{0}$
 - 2: **for** $t = 1$ to T **do**
 - 3: Update reservoir:
- $$\mathbf{h}_t = (1 - \alpha)\mathbf{h}_{t-1} + \alpha \tanh(\mathbf{W}_{in}\mathbf{x}_t + \mathbf{W}\mathbf{h}_{t-1})$$

- 4: **end for**
- 5: Compute output:

$$\mathbf{o} = \mathbf{W}_{out}\mathbf{h}_T$$

- 6: Apply softmax to obtain confidence p
- 7: $y \leftarrow \arg \max \mathbf{o}$
- 8: **return** (y, p)

The GRU output confidence is obtained via

$$p_{GRU} = \text{softmax}(\mathbf{W}_c\mathbf{h}_T + \mathbf{b}_c). \quad (16)$$

The GRU is trained using categorical cross-entropy loss

$$\mathcal{L}_{GRU} = - \sum_i y_i \log(\hat{y}_i). \quad (17)$$

F. CONFIDENCE-GUIDED CULTURAL DECISION FUSION

The final cultural confidence is computed as

$$p_{final} = \max(p_{ESN}, p_{GRU}). \quad (18)$$

Cultural enrichment is activated only if

$$p_{final} \geq \tau, \quad (19)$$

where τ is an empirically chosen threshold.

Algorithm 2 GRU Cultural Classification Network**Require:** Feature sequence $\{\mathbf{x}_t\}$ **Ensure:** Predicted class y , confidence p

- 1: **for** $t = 1$ to T **do**
- 2: Update gates:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z\mathbf{x}_t + \mathbf{U}_z\mathbf{h}_{t-1})$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r\mathbf{x}_t + \mathbf{U}_r\mathbf{h}_{t-1})$$

- 3: Compute candidate state:

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}\mathbf{x}_t + \mathbf{U}(\mathbf{r}_t \odot \mathbf{h}_{t-1}))$$

- 4: Update hidden state:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

- 5: **end for**
- 6: Apply softmax on final state
- 7: $y \leftarrow \arg \max$
- 8: **return** (y, p)

Algorithm 3 Confidence-Guided Culturally Aware Image Captioning**Require:** Input image I , confidence threshold τ **Ensure:** Final caption C_{final}

- 1: Extract visual features $\mathbf{x}$ from I
- 2: Generate base caption C_{base}
- 3: Predict cultural class and confidence using ESN:

$$(y_e, p_e) = \text{ESN}(\mathbf{x})$$

- 4: Predict cultural class and confidence using GRU:

$$(y_g, p_g) = \text{GRU}(\mathbf{x})$$

- 5: $p \leftarrow \max(p_e, p_g)$
- 6: **if** $p \geq \tau$ **then**
- 7: Enrich C_{base} with cultural context
- 8: **end if**
- 9: Refine caption linguistically
- 10: Output C_{final}

Algorithm 4 Confidence-Based Cultural Injection**Require:** Base caption C , confidence p , threshold τ **Ensure:** Updated caption C'

- 1: **if** $p \geq \tau$ **then**
- 2: Inject cultural descriptors
- 3: **else**
- 4: Preserve generic caption
- 5: **end if**
- 6: **return** C'

The selected cultural mode is

$$y^* = \begin{cases} y_{ESN}, & \text{if } p_{ESN} \geq p_{GRU}, \\ y_{GRU}, & \text{otherwise.} \end{cases} \quad (20)$$

G. CULTURAL INJECTION STRATEGY

When activated, a controlled stylistic modifier is applied

$$C_{final} = \delta(y^*) \oplus C_{base} \oplus \psi(y^*), \quad (21)$$

where $\delta(\cdot)$ and $\psi(\cdot)$ denote prefix and suffix cultural descriptors.

Injection tokens are selected based on dominant ESN feature activations, enabling explainable and bounded enrichment.

H. TRAINING CONFIGURATION

The ESN is trained using synthetic and curated captions with balanced cultural classes. The GRU is trained for 20 epochs using Adam optimization with early stopping. All inference is performed on CPU hardware. Fig. 1 illustrates the complete workflow of the proposed confidence-guided culturally aware image captioning system, highlighting parallel ESN and GRU cultural classification, confidence-based decision branching, and controlled cultural injection to suppress hallucinations under uncertainty.

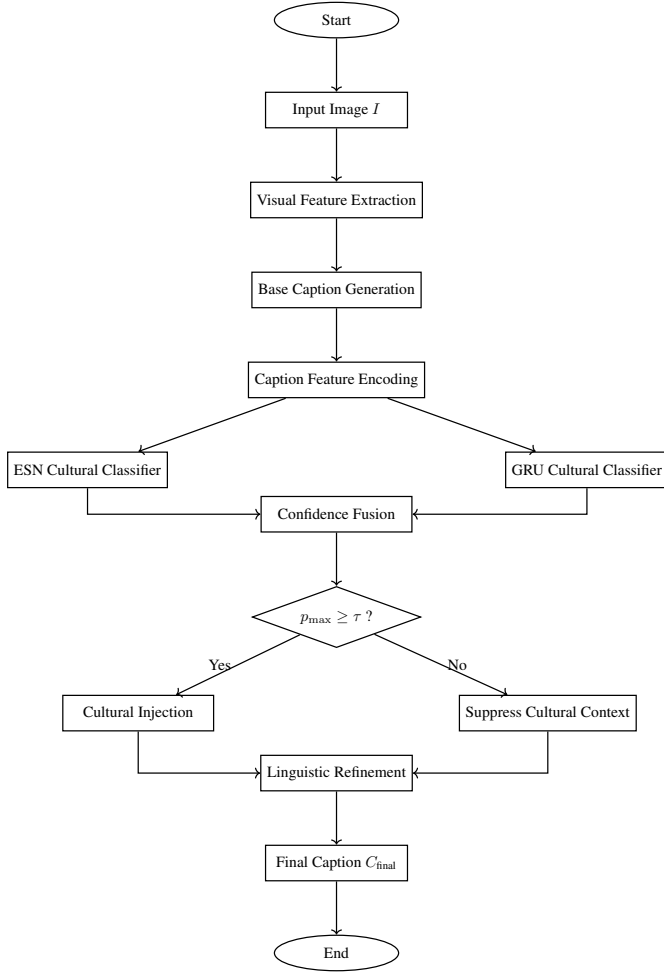


FIGURE 1: Flowchart of the proposed confidence-guided culturally aware image captioning framework.

IV. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed confidence-guided culturally aware image captioning framework. The results are discussed across qualitative system behavior, captioning quality metrics, cultural classification performance, and hyperparameter sensitivity analysis of the ESN and GRU models. All results are reported module-wise to demonstrate the effectiveness, interpretability, and robustness of the proposed system.

TABLE 2: Caption Generation Performance Metrics

Metric	Score
BLEU-1	0.5494
BLEU-2	0.4439
BLEU-3	0.3606
BLEU-4	0.2889
METEOR	0.5926
ROUGE-L	0.6733
BERTScore	0.9386
CIDEr	2.8303

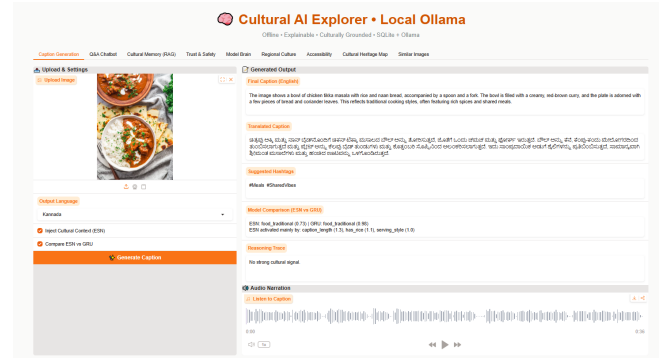


FIGURE 2: End-to-end system interface showing image upload, generated caption, multilingual translation, cultural reasoning trace, hashtag suggestions, and audio narration.

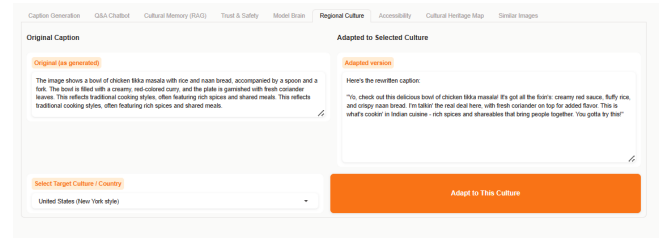


FIGURE 3: Regional culture adaptation interface showing original caption and culturally adapted caption for a selected target region.

A. END-TO-END SYSTEM OUTPUTS

Fig. 2 illustrates the complete user-facing interface of the system. The framework integrates image upload, caption generation, multilingual translation, cultural reasoning, audio narration, and model comparison within a unified workflow. The generated captions remain visually grounded, while cultural enrichment is selectively injected based on confidence, as shown by the reasoning trace panel.

B. REGIONAL CULTURAL ADAPTATION RESULTS

The regional culture module enables stylistic and cultural rewriting of captions without altering visual semantics. Fig. 3 shows an example where a base caption describing traditional Indian cuisine is adapted to a United States (New York) cultural tone, demonstrating controlled stylistic transformation rather than hallucinated content insertion.

C. CAPTIONING QUALITY METRICS

The caption generation module was evaluated using standard metrics including BLEU, METEOR, ROUGE-L, CIDEr, and BERTScore. The system achieved strong semantic alignment and linguistic fluency, confirming that confidence-based cultural gating does not degrade base caption quality. High CIDEr and BERTScore values indicate effective consensus-based relevance and contextual similarity.

TABLE 3: ESN Cultural Classification Results

Class	Precision	Recall	F1
food_traditional	0.72	0.97	0.83
festival_context	0.87	0.84	0.85
daily_life	0.79	0.50	0.61
generic	0.00	0.00	0.00

TABLE 4: GRU Cultural Classification Results

Class	Precision	Recall	F1
food_traditional	0.91	0.78	0.84
festival_context	0.83	0.69	0.75
daily_life	0.82	0.46	0.59
generic	0.57	1.00	0.73

D. ESN HYPERPARAMETER SENSITIVITY ANALYSIS

The Echo State Network was analyzed with respect to leak rate, reservoir size, and spectral radius. Validation accuracy peaks at higher leak rates, indicating improved temporal integration. Moderate reservoir sizes outperform extremely small or excessively large reservoirs, while spectral radii below unity yield stable and accurate performance, consistent with echo state theory.

E. ESN RESERVOIR DYNAMICS VISUALIZATION

To analyze internal state dynamics, PCA was applied to the ESN reservoir states. Fig. 4 shows meaningful separation for food_traditional and festival_context, while partial overlap between daily_life and generic reflects semantic similarity, highlighting the interpretability of reservoir-based cultural reasoning.

F. GRU TRAINING BEHAVIOR

The GRU-based cultural classifier was trained for 20 epochs. Fig. 6 shows stable convergence, with validation accuracy closely following training accuracy, indicating minimal overfitting and effective learning of sequential linguistic patterns.

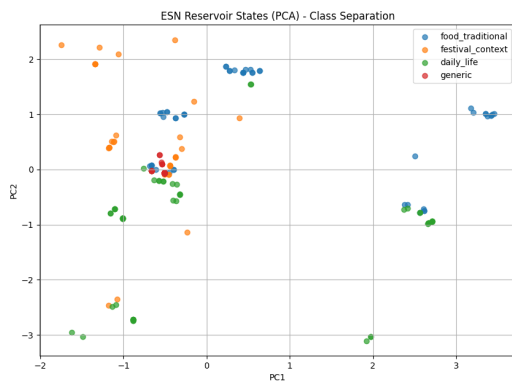


FIGURE 4: PCA visualization of ESN reservoir states showing cultural class separation.

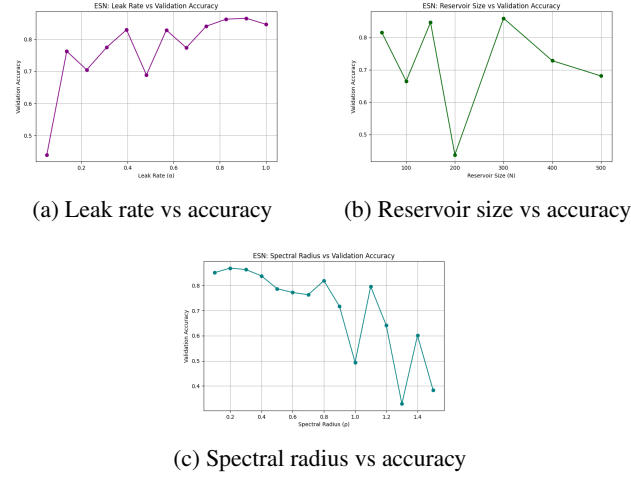


FIGURE 5: Hyperparameter sensitivity analysis of the ESN cultural classifier.

G. CULTURAL CLASSIFICATION CONFUSION ANALYSIS

Fig. 7 compares the confusion matrices of the ESN and GRU classifiers. The ESN performs strongly for food_traditional and festival_context, while showing confusion between daily_life and generic. The GRU achieves higher overall accuracy and near-perfect classification for the generic class, demonstrating stronger discriminative capacity.

H. DISCUSSION

The results demonstrate that the proposed framework effectively balances cultural richness and hallucination control through confidence-guided decision-making. The ESN provides fast, interpretable cultural signals with minimal computational overhead, while the GRU offers higher classification accuracy for complex linguistic patterns. Their fusion ensures cultural enrichment is applied only when supported by strong evidence. Unlike transformer-heavy approaches, the proposed system explicitly regulates cultural reasoning, improving reliability, transparency, and deployability in real-world applications.

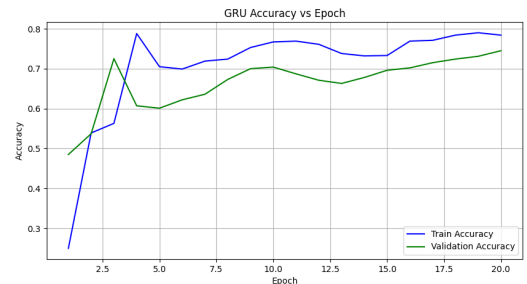


FIGURE 6: GRU training and validation accuracy across epochs.

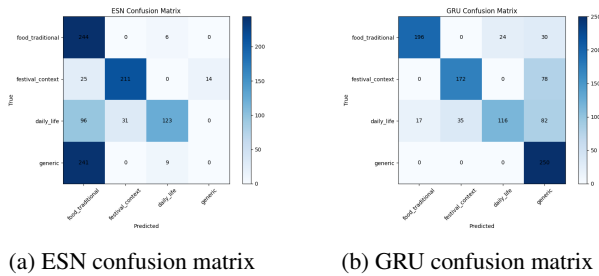


FIGURE 7: Confusion matrices comparing cultural classification performance of ESN and GRU models.

V. CONCLUSION

This work presented a confidence-guided culturally aware image captioning framework that explicitly addresses the challenge of cultural hallucination in vision–language systems. By decoupling visual caption generation from cultural reasoning and introducing a confidence-based gating mechanism, the proposed approach ensures that cultural enrichment is applied only when supported by strong linguistic and semantic evidence. The system maintains high caption quality while avoiding overconfident or culturally insensitive descriptions in ambiguous scenarios.

The integration of lightweight recurrent models proved effective for controlled cultural inference. The Echo State Network offered fast, interpretable predictions with minimal computational overhead, while the GRU achieved higher classification accuracy for complex linguistic patterns. Experimental results across captioning metrics, cultural classification performance, and confusion analysis demonstrated that the complementary strengths of ESN and GRU enable reliable and robust cultural decision-making without relying on large, opaque transformer-based cultural modules.

Overall, the proposed framework demonstrates that culturally grounded image captioning can be achieved through modular, explainable, and computationally efficient design. By combining probabilistic confidence modeling, selective cultural injection, and multilingual refinement within a unified pipeline, the system provides a practical solution for real-world deployment in culturally diverse environments where safety, interpretability, and reliability are critical.

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